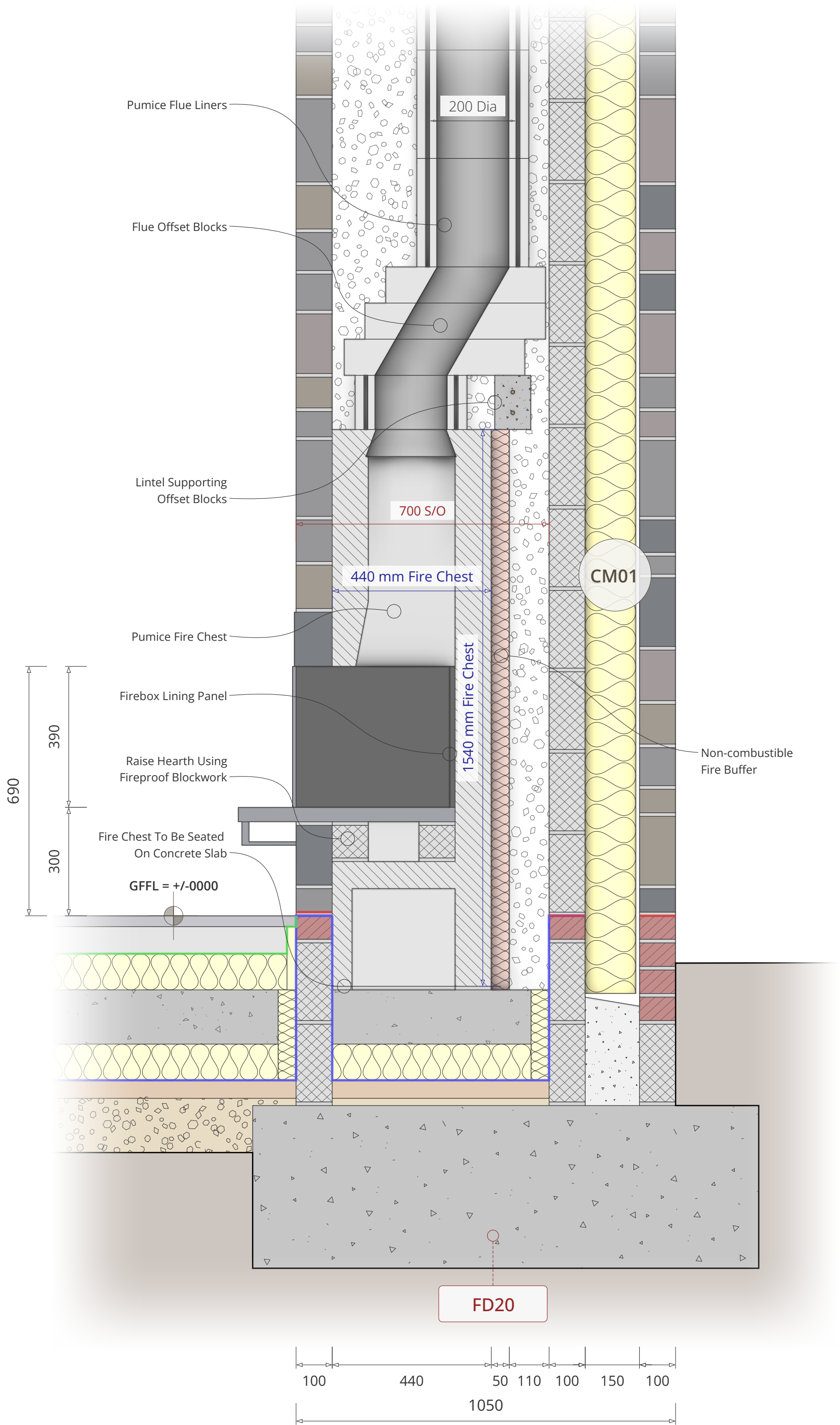




Element	Specifications
Inner Flue Liner	Construct the flue using Schiedel Isokern DM (Double Module) DM44 pumice inner liners. Component Dimensions: 200mm Internal Diameter x 310mm External Width x 300mm Standard Height per unit. Installation: Install liners 'socket up' (female socket top) to contain acidic condensate and seal all joints using Isokern Lip Glue. Ensure the liner column is fully supported at the base, typically utilizing the specific 150mm starter block or support lintel if transitioning from the Firechest gather.
Flue Offset	Encase the liners using Isokern DM44 Outer Casing blocks, staggering the joints relative to the inner liner. Create the required stack alignment 'kink' using 15° or 30° offset blocks (max 45° deviation), ensuring the annular gap between liner and casing remains strictly free of mortar droppings to preserve the insulation value. If the offset is significant, install a concrete support slab as per manufacturer details.
Terminal Fan	Install an Exodraft RSV Vertical Discharge Chimney Fan (Matte Black Cast Aluminum) to the head of the chimney stack using the proprietary flange kit. This unit creates a mechanical vacuum to overcome the offset resistance and the negative pressure of the airtight extension. Wire to a fused spur at the stack and an Exodraft EFC21 control switch located adjacent to the fireplace.
Top-Sealing Damper	Install a stainless steel Top-Sealing Damper (Lyemance or Energy Top) at the flue terminal, integrated with or positioned below the fan unit. Operated via a stainless steel cable from the hearth, this damper creates a 100% airtight seal when the fire is dormant, satisfying Part L airtightness requirements. Install a safety placard at the handle stating 'OPEN BEFORE FIRE'.
Design Rationale	The mechanical Exodraft fan is specified to overcome the negative pressure of the modern airtight extension, ensuring smoke is ejected safely despite the restrictions of the offset flue. The 'Metal Box' aesthetic is functional, housing this essential fan unit.



Element	Specifications
Fireplace Chest	Construct the fireplace recess using a Schiedel Isokern Magnum Firechest Model 500 (Compact). Product is manufactured from lightweight pumice concrete. Bond all rising components together using the supplied proprietary Isokern Lip Glue to ensure the correct aerodynamic throat angle is maintained. To achieve the client's requested 'shallow/linear' aesthetic, reduce the effective opening height by installing a raised internal hearth by using dense concrete blockwork. The Firechest must NOT sit on the floor screed. It must be bedded directly onto the Structural Concrete Slab (Constructional Hearth) using a 10-20mm 1:4 Sand/Cement mortar bed to ensure level. Do not use mortar for the upper component joints; use Lip Glue only.
Structural Void Dimensions	Builder Instruction: The Magnum 500 unit is 685mm Wide x 440mm Deep x 1540mm High. You must form the surrounding structural blockwork (CH10) to create a clear internal void of 785mm Wide x 470mm Deep. This dimension accounts for the unit size plus the mandatory 50mm expansion gap on all sides (matching CH10 requirements). CRITICAL: This gap must NOT be filled with mortar or bonded to the blockwork; pack it loosely with Rockwool or LECA to allow the pumice unit to expand thermally without cracking the surrounding masonry.
Firebox Lining	Install a 'Reeded' or 'Plain' Cast Iron Lining Panel set to the interior of the firechest. Specification: Specialist supply sized to fit the Magnum 500 opening (approx 510mm internal). Thickness: Nominal 15mm panels. Fixing: Panels must be mechanically fixed using retained lugs or high-temp anchors with a minimum 12mm expansion gap behind the plate to prevent warping. CRITICAL: Do not grout the panels tightly against the pumice shell. The panels will sit on the newly raised internal hearth base.
Throat Assembly	Install the pre-cast Pumice Throat unit directly onto the firechest side walls. Ensure the internal 'shelf' is kept clean of mortar droppings. Warning: Do not install a traditional cast iron throat damper here; it will leak air and fail the Part L test. [cite_start]Use the Top-Sealing damper specified in section CH30.
Combustion Air Supply	Install a Rytons A1 Fire-rated Periscope Underfloor Ventilator (or equivalent 150mm rigid duct system) running beneath the floor screed from an external air brick to a floor grille located within the non-combustible hearth zone immediately adjacent to the fire basket. [cite_start]Warning: This direct air feed is MANDATORY for compliance in this airtight extension; omitting it will cause the fire to starve, resulting in dangerous smoke spillage and Carbon Monoxide alarm activation.
Hearth Construction	Construct a non-combustible constructional hearth with a minimum thickness of 125mm (Structural Slab). The Firechest sits directly on this slab. CRITICAL SCREED INTERFACE: The floor screed and insulation must NOT run under the Firechest; they must abut the unit with a 10mm perimeter expansion strip (mineral wool or foam) to prevent cracking. Finish with the client's chosen 30mm Honed Black Granite or Riven Slate overlay. Ensure the hearth projects a minimum of 500mm in front of the fireplace opening and 150mm to either side.

Building Control Approval Submission

This drawing is submitted for Building Control approval. Construction must not commence until the Inspector sanctions the proposed structural details, thermal performance, and moisture integrity. The Principal Contractor is fully responsible for coordinating all statutory inspections to verify strict compliance with the Building Regulations 2010.

CM20 - Chimney Firebox Details

